Michael Madden

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Education

University of Florida | Gainesville, FL | GPA: 3.97 **August 2023 – December 2026**

- Bachelor of Science in Mechanical Engineering
- Bachelor of Science in Aerospace Engineering

Florida State University | Tallahassee, FL | GPA: 3.95 **August 2022 – August 2023**

- Associate of Arts

Work Experience

Teaching Assistant | Design and Manufacturing Lab | Gainesville, FL **January 2025 – Present**

- Supervised manufacturing labs for 100+ students, instructing safe operation of mills, lathes, and presses while ensuring the precision machining of 32 functional air engine assemblies per semester
- Designed and programmed CNC toolpaths to manufacture standardized components, reducing manufacturing time by 60% and ensuring consistency across student-built air engine assemblies
- Reviewed student design projects and technical reports, delivering feedback aligned with ANSI and ISO standards to improve engineering communication and design-for-manufacturing practices

Undergraduate Researcher | Space Technology and Research Lab | Gainesville, FL **July 2024 – Present**

- Designed a compact 56 g CubeSat reaction wheel system using magnetic levitation to test stabilization and regenerative braking
- Implemented a robotic arm platform for laboratory demonstrations of satellite servicing, enabling controlled testing of in-space servicing, assembly, and manufacturing operations, focusing on the characterization of contact dynamics
- Developed an augmented reality and camera tracking-based satellite operator interface to simplify satellite command
- Streamlined satellite control interface systems, improving operator training efficiency by 25%

Systems Engineering Intern | Space Dynamics Laboratory | Albuquerque, NM **May 2025 – July 2025**

- Designed a mission concept for twin 6U CubeSats to approach and dock, leveraging a novel interface and electric propulsion to enable on-orbit transfer of power, data, and force
- Performed feasibility studies for 3 satellite missions using structural and thermal analysis, GSE design, flight software, and subsystem development (controls, power, communications, and propulsion)
- Coordinated communication between 5 members of the Air Force Research Laboratories University Nanosat Program (UNP) and the University of Florida, ensuring alignment on mission design reviews, deliverables, and technical requirements
- Mentored and trained a 20-member UF student team in systems engineering practices to propose a CubeSat mission for the UNP NS13 nanosatellite funding cycle

Involvement

Orbital Gators | Executive Board - Program Manager | UF **April 2024 – Present**

- Founded and scaled Orbital Gators into a 5-team organization of 50+ students across multiple disciplines
- Collaborated with 150+ UF faculty members in the Astraeus Space Institute to launch NASA-funded space research initiatives
- Directed operations as Program Manager, coordinating 120+ team meetings, aligning technical milestones between teams, and ensuring completion of project deliverables
- Designed a CubeSat-scale androgynous docking interface featuring magnetic soft-capture, enabling autonomous capture from up to 25 cm and 10x greater alignment tolerance than existing satellite docking approaches
- Developed a dual robotic arm system utilizing tool changing for cost-optimized in-space servicing, assembly, and manufacturing
- Led the Bus Team, overseeing CubeSat structures development using advanced manufacturing techniques and guiding in-house development of a CubeSat attitude determination and control system

Skills and Qualifications

- SolidWorks (CSWA), Fusion 360, Abaqus
- Ansys STK Level 2 Master Certified, NASA GMAT
- Microsoft Office, G-Suite, Adobe Suite
- Arduino programming for AI and ML Algorithms
- C++, MATLAB, F’, Python, Java, HTML
- Systems engineering and requirements flowdown
- 3D printing, soldering, MIG and TIG welding
- CNC mill and lathe manufacturing